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TESTING
CNAS L0098



TEST REI

PAGE 1 OF 19

Applicant	
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Post Code:	350008
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检 验 报 告

Test Report



福建省产品质量检验研究院

Fujian Inspection and Research Institute for Product Quality (FQII)

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**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 1 OF 19

NO. (2020)DZI0021

Applicant Name: Address: Post Code: Phone / Fax: E-mail:	Fujian GIBO Kitchen & Bath tech Co.,Ltd Bld.54, Part B,Pushang Industrial Park, Fuzhou, Fujian, China 350008 18959111663 lidaliang@gibol.com.cn		
Sample(s) Name: Type/Model: Trademark: Manufacturer: Amount: Sample grade: Serial number: Production date: Date Received: State Received:	Automatic Thermostatic Faucet GBL-6108D、GBL-6101-1D、GBL-6101-2D GIBO Fujian GIBO Kitchen & Bath tech Co.,Ltd 1 pc / / / 2020-05-08 The sample has well-tended appearance before the test,and work normally when it is powered on.		
Test Specification	EN61000-6-1:2007 Electromagnetic compatibility (EMC) - Part 6-1: Generic standards - Immunity for residential, commercial and light-industrial environments EN61000-6-3:2007+A1:2011 Electromagnetic compatibility (EMC) - Part 6-3: Emission - standard for residential, commercial and light-industrial environments		
Test Results	see following pages		
Conclusion	This submitted sample was deemed to have complied with the above requirements.		
Test Date(s)	2020-05-25 to 2020-05-30	Test Location	Mawei Lab
Note			

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Approved by:
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Checked by:
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林成创

Tested by:
(signature)

陈剑文



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 2 OF 19

NO. (2020)DZI0021

Contents

1 General test information	3
1.1 General Description of Part	3
1.2 Description of operating modes	3
1.3 EUT operating conditions	3
1.4 Date of Order	3
1.5 Submitted Sample(s)	3
1.6 Test Duration	3
1.7 Support equipment	3
2 Test standards and test results summary	4
2.1 Test standards	4
2.2 Test results summary	4
3 Emissions	6
3.1 Radiated emissions on enclosure port	6
4 Immunity	9
4.1 Standard compliance/severity level/criteria	9
4.2 Performance criterion	11
4.3 Electrostatic discharges	12
4.4 Radio-frequency electromagnetic field	14
Appendix A Equipments used during test	16
Appendix B EUT test photo	17
Appendix C Photo of EUT	19



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 3 OF 19

NO. (2020)DZI0021

1 General test information

1.1 General Description of Part

Product: Automatic Thermostatic Faucet
Manufacturer: Fujian GIBO Kitchen & Bath tech Co.,Ltd
Brand Name: GIBO
Model Number: GBL-6108D
Rating: DC 6V (Four batteries in series)
Additional Model Number(s): GBL-6101-1D、GBL-6101-2D
Model Difference: Differences between the models are the name , the appearance and the decoration, do not affect the performance of EMC.
Remark:

1.2 Description of operating modes

Operating mode	Detail description
Mode 1	The normal working condition.

1.3 EUT operating conditions

The EUT exercise program used during radiated measurement was designed to exercise the various system components in a manner similar to a typical use. The standard test output as following:

I/O port type	quantity	Tested with
/	/	/

1.4 Date of Order

2020-07-07

1.5 Submitted Sample(s):

1 Sample

1.6 Test Duration

2020-05-25 to 2020-05-30

1.7 Support equipment

/



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 4 OF 19

NO. (2020)DZI0021

2 Test standards and test results summary

2.1 Test standards

EN 61000-6-1: 2007	Electromagnetic compatibility (EMC) - Part 6-1: Generic standards - Immunity for residential, commercial and light-industrial environments
EN 61000-6-3: 2007+A1:2011	Electromagnetic compatibility (EMC) - Part 6-3: Emission – standard for residential, commercial and light-industrial environments

2.2 Test results summary

Emission						
Test descriptions		Test requirements	Limit	Test results	U (dB) k=2	Remark
Radiated emissions on enclosure port	Frequency range 30MHz to 1000MHz	EN 61000-6-3: 2007+A1:2011	Class B	Pass	4.7	NOTE(4)
	Frequency range above 1000MHz		/	N/A	4.7	NOTE(15)
Conducted emissions on low voltage AC mains port	0kHz-2kHz		/	N/A	3.0	NOTE(2) NOTE(3)
	0.15MHz-30MHz		/	N/A	3.0	NOTE(4) NOTE(5)
Conducted emissions on DC power port			/	N/A	3.0	NOTE(4)
Conducted emissions on Telecommunications/network port			/	N/A	/	NOTE(4) NOTE(5) NOTE(6)
Immunity						
Test descriptions		Test requirements	Performance Criterion	Test results	U (dB) k=2	Remark
Electrostatic discharges		EN 61000-6-1: 2007	B	Pass	5%	
Radio-frequency electromagnetic field			A	Pass	1.9	NOTE(9)
Radio-frequency common mode			A	N/A	1.9	NOTE(10) NOTE(11)
Power-frequency magnetic field			A	N/A	N/A	NOTE(7) NOTE(8) NOTE(16)
Fast transients			B	N/A	10%	NOTE(11) NOTE(13)
Surges			B	N/A	10%	NOTE(12)
Voltage dips and interruptions			B/B/C	N/A	2%	NOTE(14)



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 5 OF 19

NO. (2020)DZI0021

Remarks:

- (1) If the internal emission source(s) is operating at a frequency below 9kHz the measurements need only to be performed up to 230MHz.
- (2) Applicable to apparatus covered within the scope of IEC 61000-3-2, IEC 61000-3-3 or IEC 61000-3-12.
- (3) If the apparatus is covered by the scope of IEC 61000-3-11, this may be used instead of IEC 61000-3-3.
- (4) At transitional frequencies the lower limit applies.
- (5) The limits decrease linearly with the logarithm of the frequency in the range 0.15MHz to 0.5MHz.
- (6) The current and voltage disturbance limits are derived for use with an impedance stabilization network (ISN) which presents a common mode (asymmetric mode) impedance of 150Ω to the telecommunication port under test.
- (7) Applicable only to apparatus containing devices susceptible to magnetic fields.
- (8) For CRTs, the acceptable jitter depends upon the character size and is calculated for a test level of 1A/m.
- (9) The frequency range has been selected to cover the frequencies with the highest potential risk of a disturbance.
- (10) The test level can also be defined as the equivalent current into a 150Ω load.
- (11) Applicable only to ports interfacing with cables whose total length according to the manufacturer's functional specification may exceed 3m.
- (12) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor, DC ports which are not intended to be connected to a DC distribution network are treated as signal ports.
- (13) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor, The test is applicable to DC power input ports intended to be connected permanently to cables longer than 3m.
- (14) Applicable only to input ports.
- (15) The highest internal frequency of the EUT is 8 MHz, the measurement shall only be made up to 1 GHz.
- (16) Infrared induction without Hall sensor.
- (17) N/A-Not Applicable).



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 6 OF 19

NO. (2020)DZI0021

3 Emissions

3.1 Radiated emissions on enclosure port

Standard:	EN 61000-6-3
Classification of EUT:	Class B
Test result:	Pass

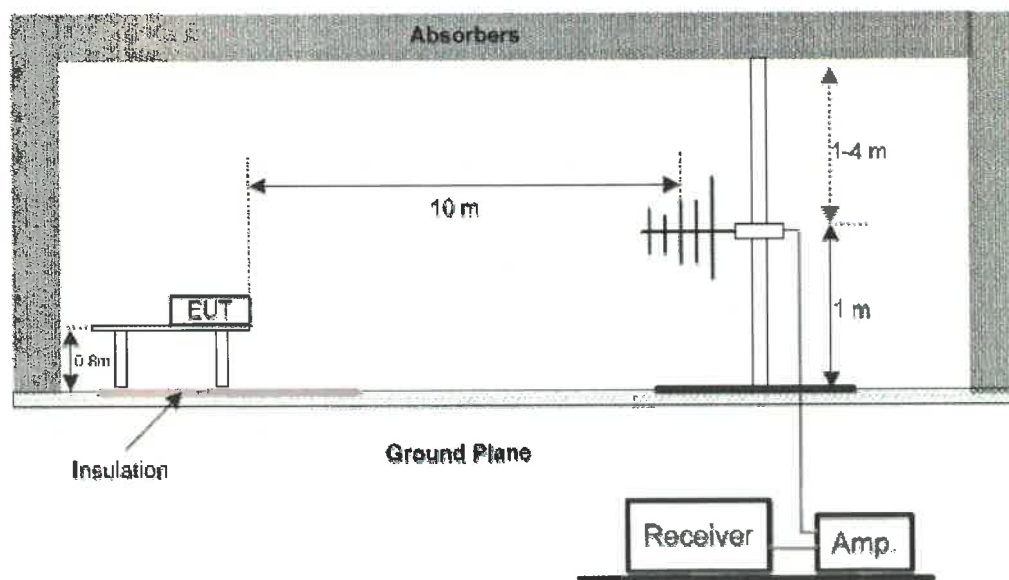
Temperature:	(17~23)°C
Relative humidity:	(50~60)%
Test date:	2020-05-25

Test voltage:	DC 6V
Operating mode:	Mode 1

3.1.1. Test method

The test was performed in accordance with EN 61000-6-3. The prescans were performed by a peak measuring receiver. The high values found were measured by a quasi-peak measuring receiver.

3.1.2. Test setup





**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 7 OF 19

NO. (2020)DZI0021

3.1.3. Limits Requirements for radiated emissions on enclosure port at frequencies up to 1 GHz

Table clause	Frequency range MHz	Measurement			Class A limits dB(μV/m)
		Facility (see Table A.1)	Distance m	Detector type/ bandwidth	
Table 1	30 to 230	OATS/SAC	10	Quasi Peak/ 120kHz	30
	230 to 1000				37
	30 to 230	OATS/SAC	3		40
	230 to 1000				47
At transitional frequencies the lower limit applies.					



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

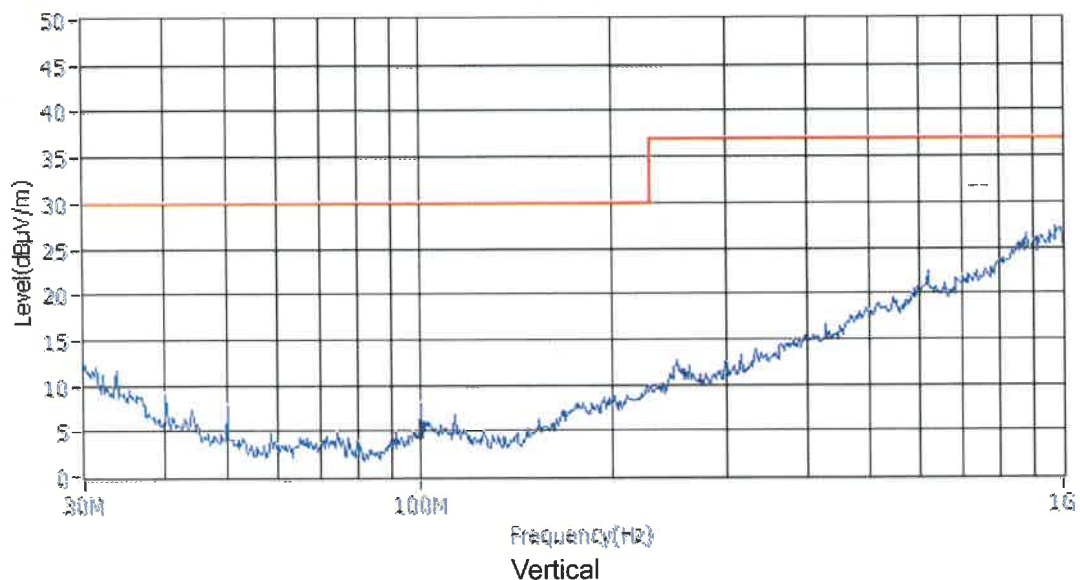
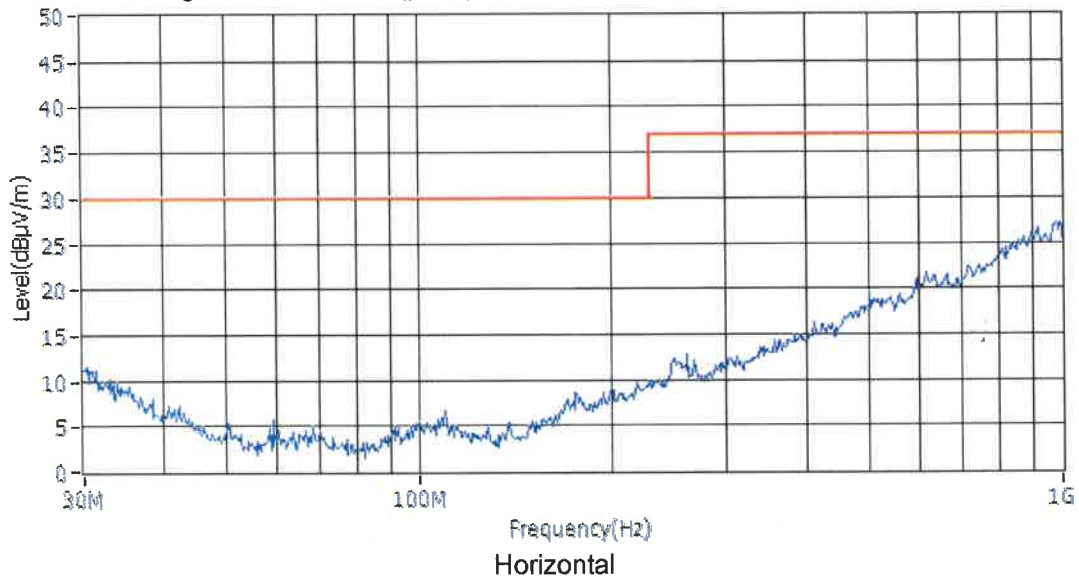
PAGE 8 OF 19

NO. (2020)DZI0021

3.1.4. Test results at 10m of SAC

Model Name	GBL-6108D	Test Voltage	DC 6V
Frequency MHz	Polarity	Limit dB(μ V/m)	Measured level dB(μ V/m)
30.0000	Vertical	30	9.4
33.7000	Vertical	30	8.9
112.8500	Vertical	30	4.1
251.1500	Vertical	37	9.8
261.4000	Horizontal	37	9.9
617.8500	Vertical	37	19.5

The reference diagrams as follows: (peak)





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 9 OF 19

NO. (2020)DZI0021

4 Immunity

4.1 Standard compliance/severity level/criteria

Clause	Tests standard No.	Test specification level/Test mode	Test ports	Criteria
Table 1 1.1	Power frequency magnetic field immunity EN 61000-4-8	50Hz or 60Hz, 3A/m(r.m.s) μ s	Enclosure	A
Table 1 1.2	Radio-frequency electromagnetic field EN 61000-4-3	80MHz to 1000MHz 3V/m(unmodulated, r.m.s), 1kHz,80% AM modulated	Enclosure	A
Table 1 1.3		1,4GHz to 2,0GMHz 3V/m(unmodulated, r.m.s), 1kHz,80% AM modulated	Enclosure	A
Table 1 1.4		2,0GHz to 2,7GMHz 1V/m(unmodulated, r.m.s), 1kHz,80% AM modulated	Enclosure	A
Table 1 1.5	Electrostatic discharge EN 61000-4-2	± 8 kV air discharge ± 4 kV contact discharge (Direct Mode)	Enclosure	B
		± 4 kV HCP discharge ± 4 kV VCP discharge (Indirect Mode)	Enclosure	B
Table 2 2.1	Radio-frequency common mode EN 61000-4-6	0.15MHz to 80MHz 3V(unmodulated, r.m.s), 1kHz 80%, AM 150ohm source impedance	Signal ports (NOTE 1) (NOTE 2)	A
Table 3 3.1			Input and output DC power ports (NOTE 1) (NOTE 2)	A
Table 4 4.1			Input and output AC power ports (NOTE 1)	A
Table 2 2.2	Electrical fast transient/burst immunity EN 61000-4-4	± 0.5 kV(peak) 5/50ns Tr/Th 5kHz Repetition Frequency	Signal ports (NOTE 2)	B
Table 3 3.3		± 0.5 kV(peak) 5/50ns Tr/Th 5kHz Repetition Frequency	Input and output DC power ports (NOTE 3)	B
Table 4 4.5		± 1 kV(peak) 5/50ns Tr/Th 5kHz Repetition Frequency	Input and output AC power ports	B



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 10 OF 19

NO. (2020)DZI0021

Clause	Tests standard No.	Test specification level/Test mode	Test ports	Criteria
Table 3 3.2	Surges EN 61000-4-5	Line to reference ground for each individual line: ± 0.5 kV 1.2/50(8/20) Tr/Th μ s	Input and output DC power ports (NOTE 4)	B
Table 4 4.4		± 1 kV (peak) 1.2/50(8/20) Tr/Th μ s (line to line) ± 2 kV (peak) 1.2/50(8/20) Tr/Th μ s (line to earth or ground)	Input and output AC power ports	B
Table 4 4.2 4.3	Voltage dips EN 61000-4-11	Residual voltage 0% 0.5 period Residual voltage 70% 25/30 periods at 50/60Hz Residual voltage 0% 1 periods	AC power ports (NOTE 5)	B B C
	Voltage interruptions EN 61000-4-11	Residual voltage 0% 250/300 periods at 50/60Hz	AC power ports (NOTE 5)	C

NOTE:

- 1) The test level can also be defined as the equivalent current into a 150 Ω load. Applicable only to ports which, according to the manufacturer's specification, support cable lengths greater than 3m.
- 2) Applicable only to ports which, according to the manufacturer's specification, may connect directly to outdoor cables.
- 3) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor. The test is applicable to DC power input ports intended to be connected permanently to cables longer than 3m.
- 4) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor. DC ports which are not intended to be connected to a DC distribution network are treated as signal ports.
- 5) Applicable only to input ports.



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 11 OF 19

NO. (2020)DZI0021

4.2 Performance criterion

Criterion A	The apparatus shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.
Criterion B	The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.
Criterion C	Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 12 OF 19

NO. (2020)DZI0021

4.3 Electrostatic discharges

Standard: EN 61000-6-1
Test result: Pass
Severity: $\pm 4\text{kV}$ for Direct & Indirect Contact Discharge
 $\pm 8\text{kV}$ for Air Discharge

Performance criterion: B

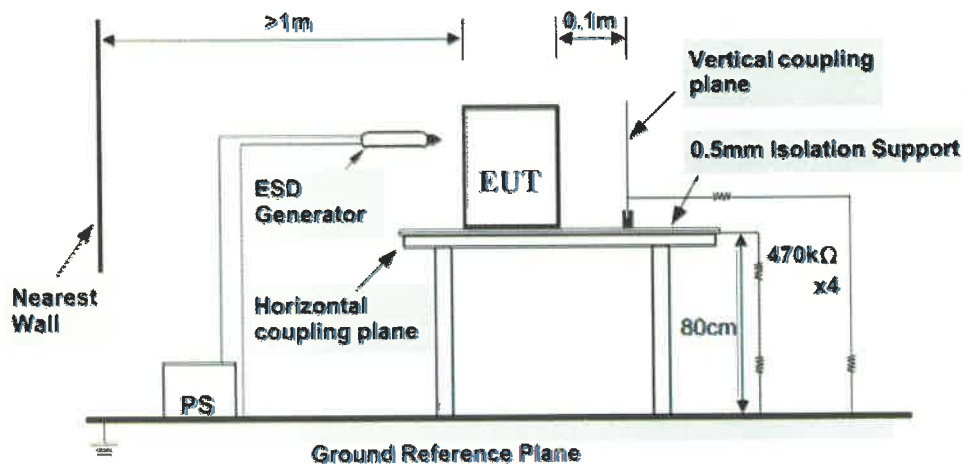
Temperature: $(17\sim 23)^\circ\text{C}$
Relative humidity: $(50\sim 60)\%$
Test date: 2020-05-26

Test voltage: DC 6V
Operating mode: Mode 1

4.3.1 Test method

The test was performed in accordance with EN 61000-4-2.

4.3.2 Test setup





**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 13 OF 19

NO. (2020)DZI0021

4.3.3 Test Result

Model Name	GBL-6108D	Test Voltage	DC 6V
Location	Discharge method	Test voltage kV	Individual results
HCP (Horizontal coupling plane)	Indirect contact	±4	B
VCP (Vertical coupling plane)	Indirect contact	±4	B
Metal part & Screws	Direct contact	±4	B
Gap	Air	±8	B
insulating surfaces	Air	±8	B

Note:1).N/A-denotes test is not applicable in device.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 14 OF 19

NO. (2020)DZI0021

4.4 Radio-frequency electromagnetic field

Standard: EN 61000-6-1
Test result: Pass
Severity: 3V/m for 80 MHz to 1000 MHz
3V/m for 1,4 GHz to 2,0 GHz
1V/m for 2,0 GHz to 2,7 GHz

Performance criterion: A

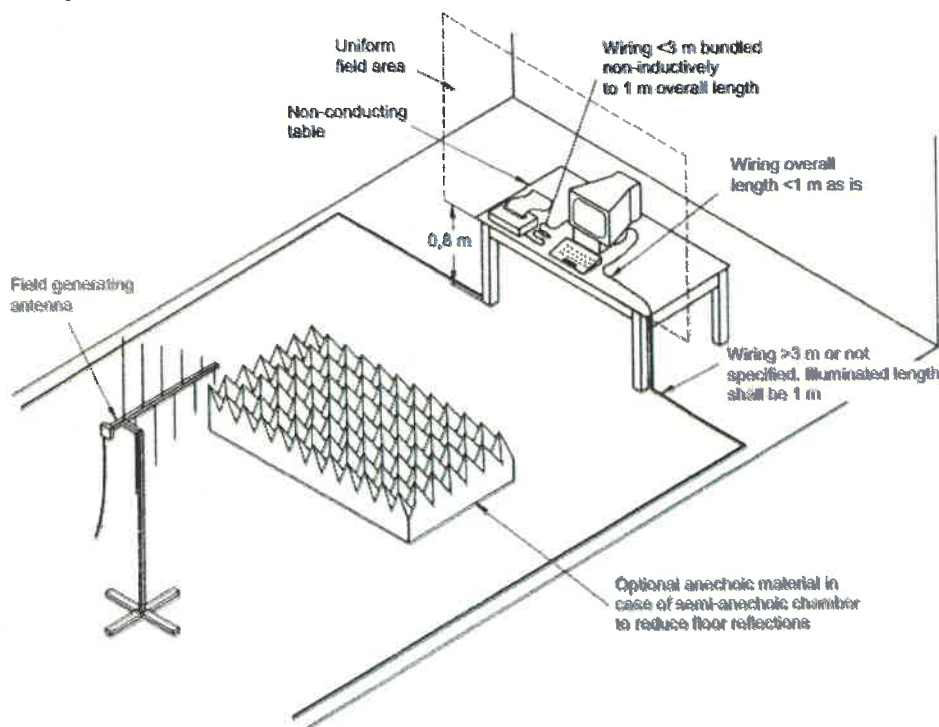
Temperature: (17~23) °C
Relative humidity: (50~60)%
Test date: 2020-05-26

Test voltage: DC 6V
Operating mode: Mode 1

4.4.1 Test method

The test was performed in accordance with EN 61000-4-3.

4.4.2 Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 15 OF 19

NO. (2020)DZI0021

4.4.3 Test Result

Model Name	GBL-6108D		Test Voltage	DC 6V		
Frequency MHz	Field strength V/m	Dwell time s	Frequency step	Polarity	Face	Individual results
80~1000	3	3	1%	Horizontal	0°	A
					90°	A
					180°	A
					270°	A
				Vertical	0°	A
					90°	A
					180°	A
					270°	A
1400~2000	3	3	1%	Horizontal	0°	A
					90°	A
					180°	A
					270°	A
				Vertical	0°	A
					90°	A
					180°	A
					270°	A
2000~2700	1	3	1%	Horizontal	0°	A
					90°	A
					180°	A
					270°	A
				Vertical	0°	A
					90°	A
					180°	A
					270°	A

Note:1).N/A-denotes test is not applicable in device.



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 16 OF 19

NO. (2020)DZI0021

Appendix A Equipments used during test

DESCRIPTION	MANUFACTURER	MODEL NO./TYPE	Management NO.	NEXT CAL
Semi-Anechoic Chamber	Frankonia	22.28m×15.98m×9.6m	DZ-25-108-1	2025-03-23
EMI Test Receiver	R&S	ESU	DZ-25-105-1	2021-03-16
Biconilog Antenna	ETS-LINDGREN	3142D	DZ-25-115	2021-06-28
Pre-Amplifier	R/S	SCU-01	DZ-25-105-3	2021-04-11
ESD Simulator	EMTEST	NX30	DZ-25-203	2020-11-01
Fully anechoic chamber	ETS Lindgren	8m×4m×3.8m	DZ-25-043-2	2021-03-01
RS Test System	R&S	TS9981	DZ-25-134	2021-03-10
Turntable	Sunol Sciences	FM1511	DZ-25-048	N/A
Turntable/ Antenna Positioning Tower	Sunol Sciences	SC99V	DZ-25-040	N/A
NOTE N/A - not applicable				



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

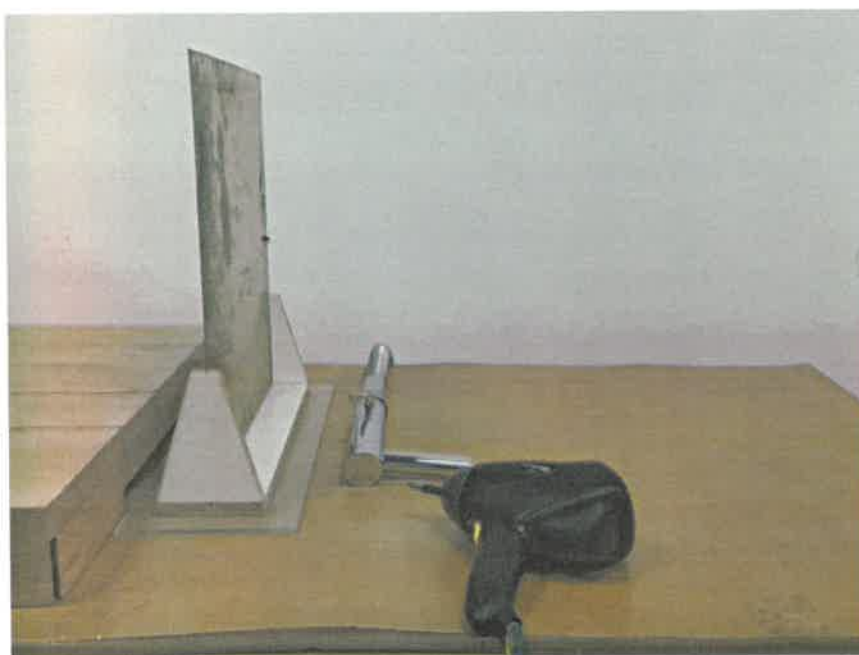
PAGE 17 OF 19

NO. (2020)DZI0021

Appendix B EUT test photo



Radiated emissions on enclosure port



Electrostatic discharges



**Fujian Inspection and Research Institute
for Product Quality**

TEST REPORT

PAGE 18 OF 19

NO. (2020)DZI0021



Radio-frequency electromagnetic field

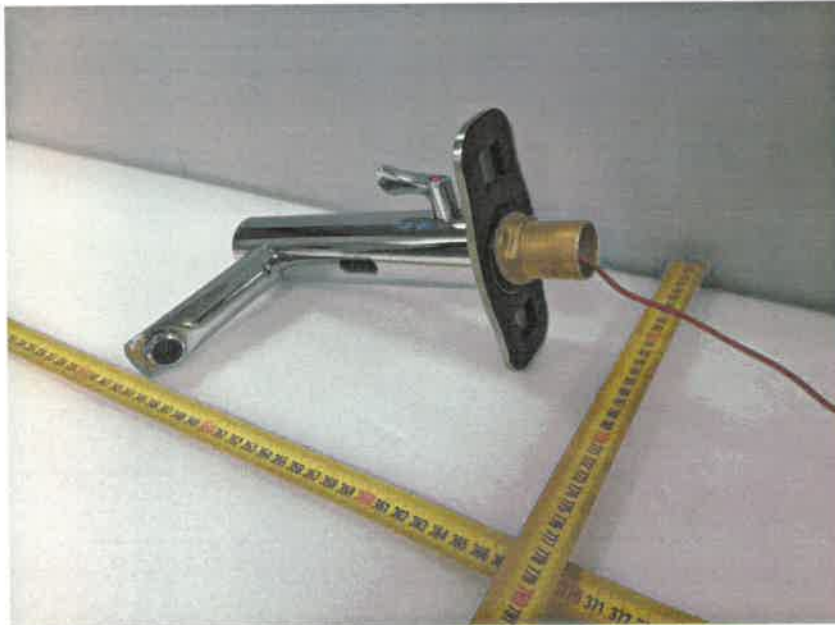


**Fujian Inspection and Research Institute
for Product Quality**
TEST REPORT

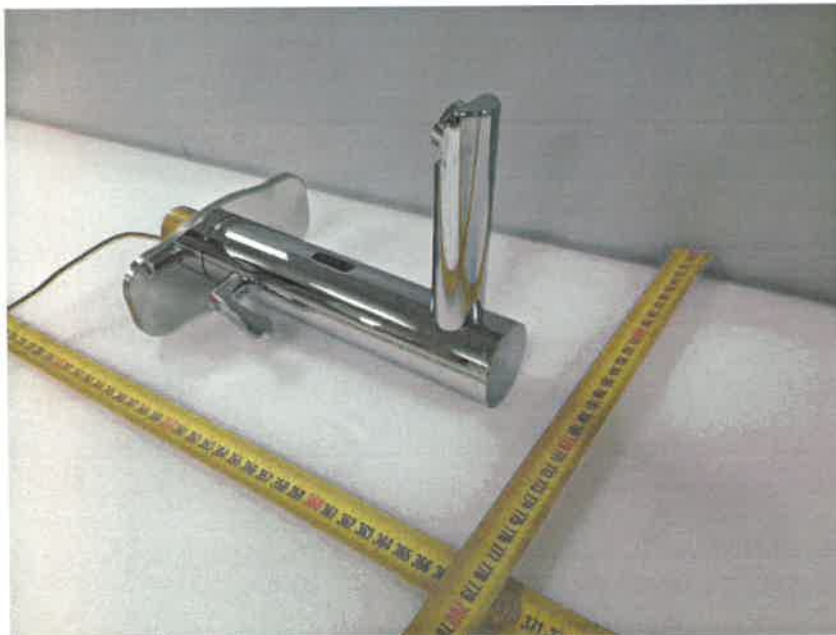
PAGE 19 OF 19

NO. (2020)DZI0021

Appendix C Photo of EUT



GBL-6108D



GBL-6108D

--End of report--

